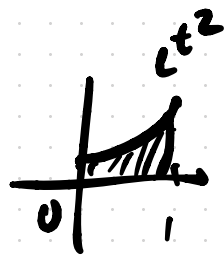


$$\int_0^1 e^{t^2} dt = \lim_{n \rightarrow \infty} \sum_{i=1}^n e^{(t_i^*)^2} \Delta t$$

100 intervals, midpoint rule

1.4626..

number/
constant



$$\int_0^x e^{t^2} dt$$

antiderivative
of e^{x^2}

$$x=0$$

$$0$$

$$x=1$$

$$1.4626\dots$$

$$x=2$$

$$16.449\dots$$

e^{x^2} has
no "elementary"
antiderivative

function of x

$$\int_0^1 t^2 dt = F(1) - F(0)$$

where F is any antiderivative
of t^2

$$= \frac{1}{3} t^3 \Big|_0^1$$

$$= \frac{1}{3} - 0 = \frac{1}{3}$$

number

$$\int_0^x t^2 dt = F(x) - F(0)$$

$$= \frac{1}{3} t^3 \Big|_0^x$$

$$= \frac{1}{3} x^3$$

function of x

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

This can be used to compute definite integrals, provided that we can write an expression for an antiderivative of f .

$$\int_{\pi/4}^{\pi/2} \sin x \, dx = F\left(\frac{\pi}{2}\right) - F\left(\frac{\pi}{4}\right)$$

where F is any
antiderivative of
 $\sin x$

$$= -\cos x \Big|_{\pi/4}^{\pi/2}$$

$$= -0 - \left(-\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\sqrt{2}}{2}$$

$$\int_7^{10} (3e^x + \frac{1}{x}) dx = (3e^x + \ln x) \Big|_7^{10}$$

$$= (3e^{10} + \ln 10) - (3e^7 + \ln 7)$$

$$= 3e^{10} - 3e^7 + \ln \frac{10}{7}$$

$$\int_0^{\pi/4} \sec^2 \theta d\theta = \tan \theta \Big|_0^{\pi/4}$$

$$= 1 - 0$$

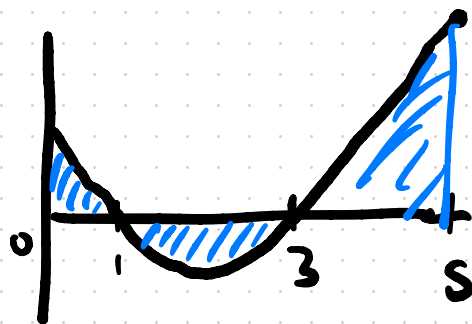
$$= 1$$

$$\int_0^5 |x^2 - 4x + 3| dx$$

$$\int_0^1 (x^2 - 4x + 3) dx$$

$$+ \int_1^3 -(x^2 - 4x + 3) dx$$

$$+ \int_3^5 (x^2 - 4x + 3) dx$$



Want sum
of blue
areas

+ - +

$$x^2 - 4x + 3 = 0$$

$$(x-1)(x-3) = 0$$

$$x = 1 \text{ or } 3$$

$$\int_0^1 (x^2 - 4x + 3) dx = \left. \frac{1}{3}x^3 - 2x^2 + 3x \right|_0^1$$
$$= \frac{1}{3}$$

$$- \int_1^3 (x^2 - 4x + 3) dx = - \left. \left(\frac{1}{3}x^3 - 2x^2 + 3x \right) \right|_1^3$$
$$= - \left(0 - \frac{1}{3} \right) = \frac{1}{3}$$

$$\int_3^5 (x^2 - 4x + 3) dx = \left. \left(\frac{1}{3}x^3 - 2x^2 + 3x \right) \right|_3^5$$

$$= \frac{20}{3} - 0 = \frac{20}{3}$$

$$\frac{4}{3} + \frac{4}{3} + \frac{20}{3} = \boxed{\frac{28}{3}}$$

Fundamental theorem of calculus, part 1

Let f be a continuous function defined on $[a, b]$, and let

$$g(x) = \int_a^x f(t) dt$$

for $a \leq x \leq b$. Then $g'(x) = f(x)$ for all $a < x < b$.

In other words, g is an antiderivative of f .

In particular, every continuous function has an antiderivative.

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

Indefinite Integrals and the Net Change Theorem (5.4)

"Indefinite integral" is a traditional name for antiderivative. In this class, we'll use the following definition.

Given a function f defined on an interval, the **indefinite integral** of f is the **general form** of an antiderivative of f on that interval. It is denoted

$$\int f(x) dx$$

In other words, it is the family of all antiderivatives of f .

$$\int (x^2 + \sqrt{x} + 2\cos x) dx$$

$$= \frac{1}{3}x^3 + \frac{2}{3}x^{3/2} + 2\sin x + C$$

a, b constants

Definite integral $\int_a^b f(t) dt$

number

Definite integral $\int_a^x f(t) dt$

function of x

Indefinite integral $\int f(t) dt$

family of
functions of t

Net change theorem

For any function F which is differentiable on (a, b) ,

$$\int_a^b F'(x) dx = F(b) - F(a)$$

This is the same as the Fundamental Theorem of Calculus.

Let $P(t)$ be the population of a bacteria colony at time t .
Suppose

$$\frac{dP}{dt} = Ae^{ct}$$

where A and c are constants. How much does the population increase by between time $t=0$ and $t=10$?

$$P(10) - P(0) = \int_0^{10} \frac{dP}{dt} dt$$

$$= \int_0^{10} Ae^{ct} dt$$

$$= \frac{A}{c} e^{ct} \Big|_0^{10}$$

$$= \frac{A}{c} (e^{10c} - 1)$$

A particle moves along a line with velocity

$$v(t) = t^2 - 4t + 3$$

What is the **displacement** of the particle between $t = 0$ and $t = 5$?

Displacement is the difference in position between the end time and start time.

position

$$s(5) - s(0) = \int_0^5 s'(t) dt$$

$$= \int_0^5 (t^2 - 4t + 3) dt$$

$$= \left. \frac{1}{3}t^3 - 2t^2 + 3t \right|_0^5$$

$$= \frac{20}{3}$$

What is the **distance** traveled by the particle between $t=0$ and $t=5$?

$$\int_0^5 |t^2 - 4t + 3| dt = \frac{28}{3}$$